

Only the selector learns

Selection drift measured directly
identity (Jaccard) · utility (ΔU)

Whole-
slide
image



Frozen CONCH

encoder ❄️
patch features
 $\{x_i\}$

Tissue
grouping

8 fixed
prompts ❄️



Hierarchical Selector (learned)

Group
scoring

Budget
allocation

Patch
scoring
+ top- b_j

Group $\rightarrow$ Patch · $\sum_j b_j = B = 8$

Selected
evidence

$|P| = B = 8$



hard budgeted evidence

Frozen
diagnosis ❄️

pool selected
evidence
cosine with
fixed $\{t_c\}$

Diagnosis
 $p(y | P)$

inference
ends here

Continual preservation during training
(training only)

Selection Memory

past selector
snapshots
scores · candidate
indices · utilities

no images, no features

Distill old
scores

(behavioral
continuity)
 L_{KD}

Preserve
utility

(functional
continuity)
 L_{eq}

Replay old
samples

(task
competence)
 L_{rep}

Task-local
LoRA update

fresh ΔW_τ ·
fold into
shared selector

One shared
selector

(used at
test time)

Preserve usefulness, not exact identity
allow different evidence if $U_{new} \geq U_{old}$

❄️ frozen (no training) · orange = learned · dashed = training only · solid path = training and inference